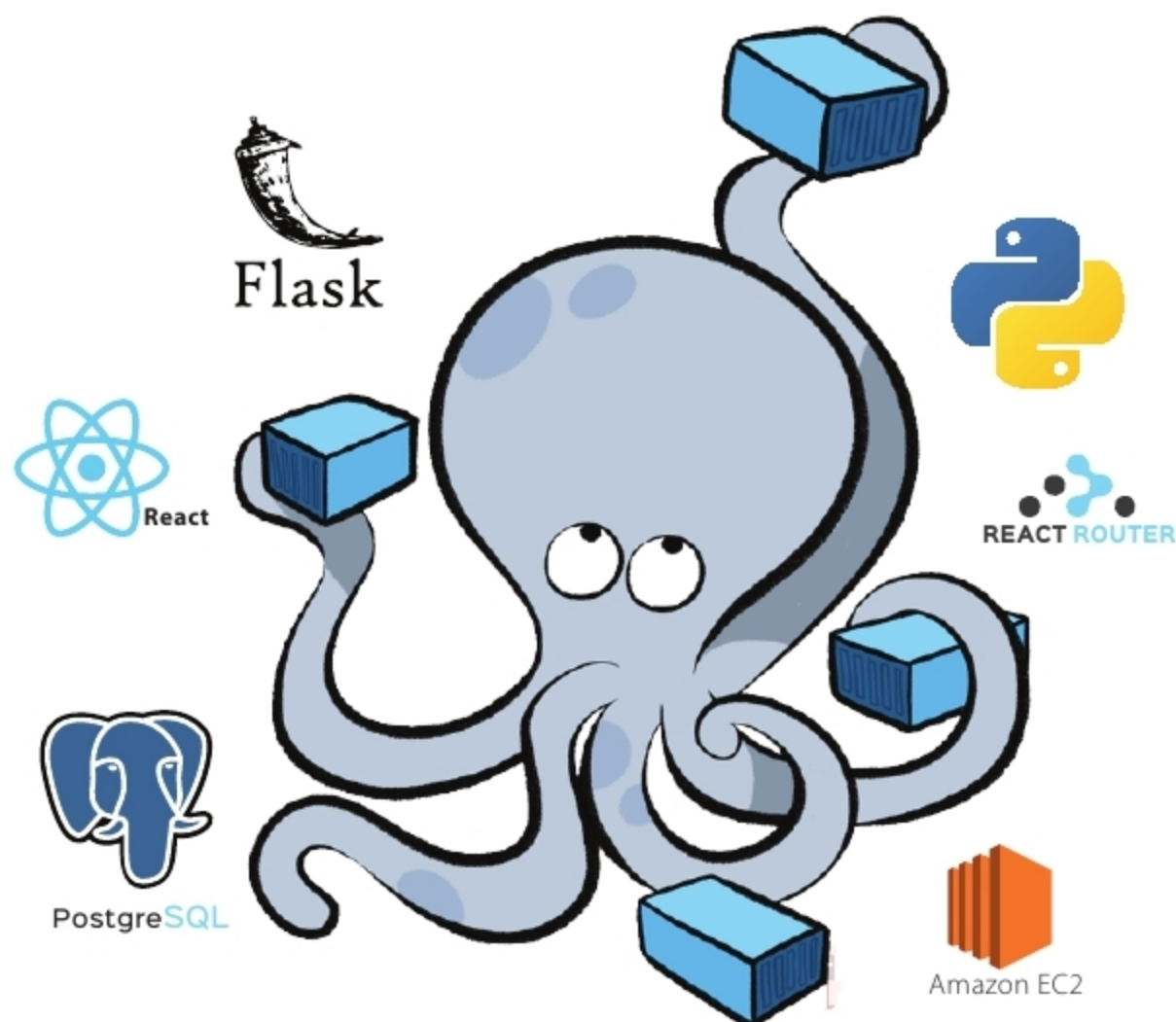


How to Run Multiple Services Inside a Single Container Using Supervisord

Supervisord or Supervisor daemon is an open source process management system. It is a client/ server system that enables the user to control numerous services. If a process crashes for any reason, Supervisor simply restarts it.

This article explains how to run multiple services inside a single container, and how to effectively use Docker Compose and persistent volume in a local development environment with Supervisord.



A container is a lightweight platform for running applications along with their dependencies in an isolated environment. It is good to run a single service inside a container.

Though we can access the various services hosted in different containers using a container network, we can get the same benefits by running multiple services in a single container. There are some situations in which we need to run more than one service inside a container, and these services should be accessible from the container host or network — for example, the Apache/ Nginx HTTP server along with the FTP

server, or some microservices that are running inside the same container with different processes.

We can use different methods such as adding multiple commands in ENTRYPOINT at the end of Dockerfile, writing a wrapper script to spawn multiple services in a container with unique process ids. But the method has some limitations such as increasing the complexity when the number of services that have to be managed in a container increases. There can also be dependency issues while starting multiple services.

We can use a simple and user friendly process manager like Supervisord to start, stop and restart

multiple services. Supervisord is a lightweight tool that can be easily deployed in a container. Its configuration is easy to maintain as the number of services is handled by Supervisor, which can be accessed by a Web interface, XML RPC interface and command line interface.

I am going to deploy a simple multi-service application using Docker Compose. Make sure the Docker daemon and Docker Compose are installed and running in your machine. Docker Compose is a CLI tool that helps you to define and deploy a multi-service containerised application environment. It is ideal for developing